

Internship Final Report

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University: Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology

Major: Data Science

Internship Duration: June 1st, 2026 - June 30th, 2026

Company: ShadowFox

Domain: Data Science

Mentor: Mr. Hariharan

Coordinator: Mr. Aakash

Objectives

My primary objectives for this internship were to:

1. Gain practical experience in Data Science and data visualization.
2. Learn how to analyze real-world datasets and extract meaningful insights.
3. Improve my skills in Python libraries such as Pandas, Matplotlib, Seaborn, and Streamlit.
4. Develop an interactive dashboard for data analysis and visualization.

Tasks and Responsibilities

During my internship, I completed the following tasks:

- **Data Visualization:** Worked with Matplotlib and Seaborn to create multiple visualizations such as line charts, bar charts, histograms, scatter plots, pie charts, box plots, and heatmaps.
- **Data Analysis:** Performed AQI data analysis using a Delhi air quality dataset containing pollutant measurements and AQI categories.

- **Dashboard Development:** Built an interactive Delhi Air Quality Analysis Dashboard using Streamlit to visualize AQI trends, station-wise analysis, seasonal AQI changes, and pollutant correlations.
- **Data Preprocessing:** Handled dataset loading, data conversion, and organization for analysis.
- **Report Documentation:** Prepared detailed reports for the completed tasks and documented findings.

Learning Outcomes

- **Technical Skills:** Improved my knowledge in Python libraries such as Pandas, Matplotlib, Seaborn, and Streamlit.
- **Data Visualization Knowledge:** Learned how to create meaningful graphs and heatmaps to represent data clearly.
- **Dashboard Development:** Understood how to build interactive dashboards and improve user experience with proper design.
- **Data Analysis Skills:** Enhanced my ability to interpret trends, patterns, and pollutant relationships in real-world AQI data.
- **Professional Development:** Improved my ability to manage tasks, document work, and complete projects within deadlines.

Challenges and Solutions

- **Understanding New Tools:** Initially, learning Streamlit and organizing the dashboard structure was challenging. I solved this by practicing and experimenting with different layouts.
- **Data Handling:** Managing large AQI datasets and analyzing multiple pollutants required careful preprocessing and grouping. This was solved using Pandas functions.
- **Visualization Design:** Adjusting graph sizes, layouts, and colors to make the dashboard look clean and user-friendly was challenging. This was improved through multiple modifications and testing.

Conclusion

My internship at ShadowFox provided valuable practical experience in Data Science. Through data visualization, AQI analysis, and dashboard development, I improved both my technical and analytical skills. This internship strengthened my understanding of real-world data science applications and increased my confidence in building practical projects.

Acknowledgments

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